Problem Chosen	2023 MCM/ICM Summary Sheet	Team Control Number
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Assessment and analysis of SDGs based on relational networks

Summary

The Sustainable Development Goals (SDGs) represent a widely acknowledged and globally implemented framework for promoting development that fulfills current needs while safeguarding future generations. Given the interdependent nature of the SDGs, it is critical to investigate their mutual interactions and establish priorities for their attainment. The identification of such priorities has the potential to expedite progress towards the SDGs, ultimately improving the well-being of countless individuals worldwide. In this work, we employ a network-based approach to model the SDGs and put forth two prioritization frameworks.

In Requirement 1, the correlation impact coefficients between SDGs were obtained by processing the SDG index score data for 2021 for each country in the world using Pearson correlation analysis method. Based on the correlation impact coefficients, the relationships between SDGs were classified into three categories, synergistic, trade-offs and non-significant relationships, and thus a network of relationships between SDGs was constructed.

In Requirement 2, to determine the priorities, we designed two methods, SEM model and MCA framework. SEM model determines the priority of the goals by determining the weight of their impact on the whole system through the deep relationship between goals and indicators. MCA framework integrates two dimensions of urgency and system impact to get the priority of SDGs. Finally, the neural network simulation is used to predict the scores of the world SDGs indicators for the next ten years after the priorities are launched, and to compare the goals that can be reasonably achieved in ten years.

In Requirement 3, we obtain a new network structure based on a gray prediction model simulation to predict the changes after the realization of SDG4, and use the MCA framework to identify priorities through the interactions between SDGs. New SDGs worthy of consideration are also proposed.

In Requirement 4, we develop rules for the impact of technological progress, climate change, and regional wars on the SDGs, and modify them to simulate the occurrence of these changes based on the 2021 data. The new network structure was obtained and prioritized using the previously designed model, which led to an analysis of the impact on the UN's progress toward the SDGs.

In Requirement 5, we developed a three-step framework for companies and organizations to analyze industry value chains, collect metrics data, and identify corporate priorities to help them prioritize their goals.

Keywords：Relationship Network　Pearson correlation analysis　SEM Model　Multi-criteria analysis　Neural Networks　Gray prediction model　SDG

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1 Introduction

1.1 Problem Background

In order to address the many challenges facing humanity today, such as environmental degradation, poverty, and inequality, and to improve the lives and future of everyone, the United Nations established 17 Sustainable Development Goals in 2015, requiring collective efforts to build an inclusive, sustainable, and resilient future for humanity and the planet. To achieve sustainable development, the three core elements of economic growth, social inclusion, and environmental protection must be coordinated. These factors cover all aspects of personal and social life, and interact with each other in complex ways. Therefore, actions toward one goal may strengthen or offset actions toward another goal, leading to synergy or balance between the sustainable development goals, making the achievement of the goals an uncertain process. In addition, unpredictable factors such as technological advances, war conflicts, and global pandemics will have significant impacts on achieving these goals.

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Create a network of the relationships between 17 SGDs
- Identify priorities that can most effectively advance the work of the UN and evaluate the effectiveness. What can be achieved 10 years after priorities are initiated.
- How the network structure changes after a goal is achieved, how the priorities set will affect, and whether new goals need to be adopted by the United Nations
- The impact of technological progress, climate change or other international crises on networks and the choice of priorities. What is the impact on the progress of the United Nations from a network perspective?
- How the proposed network approach determines target priorities in other domains

1.2 Our Work

We performed a Pearson correlation analysis on the scores of all goals and selected indicators from diverse nations, leading to the construction of a network illustrating the relationships among the 17 goals as well as between the goals and the varied indicators based on the correlation coefficients. Additionally, two models, namely MCA and SEM, were employed to conduct a priority analysis based on the correlation coefficients.

Upon initiation of certain priority items, we employed a BP neural network to predict the target score data for the upcoming decade. In instances where priority items were already implemented, we utilized the grey forecasting model to predict other data post-implementation and conducted a secondary analysis of the associated data to construct the relationship network and priority.

In cases of significant international events, we established a set of rules for changes in target score data through comprehensive assessments, and analyzed the impact of such events on the relationship network and priority of the goals based on the modified data.

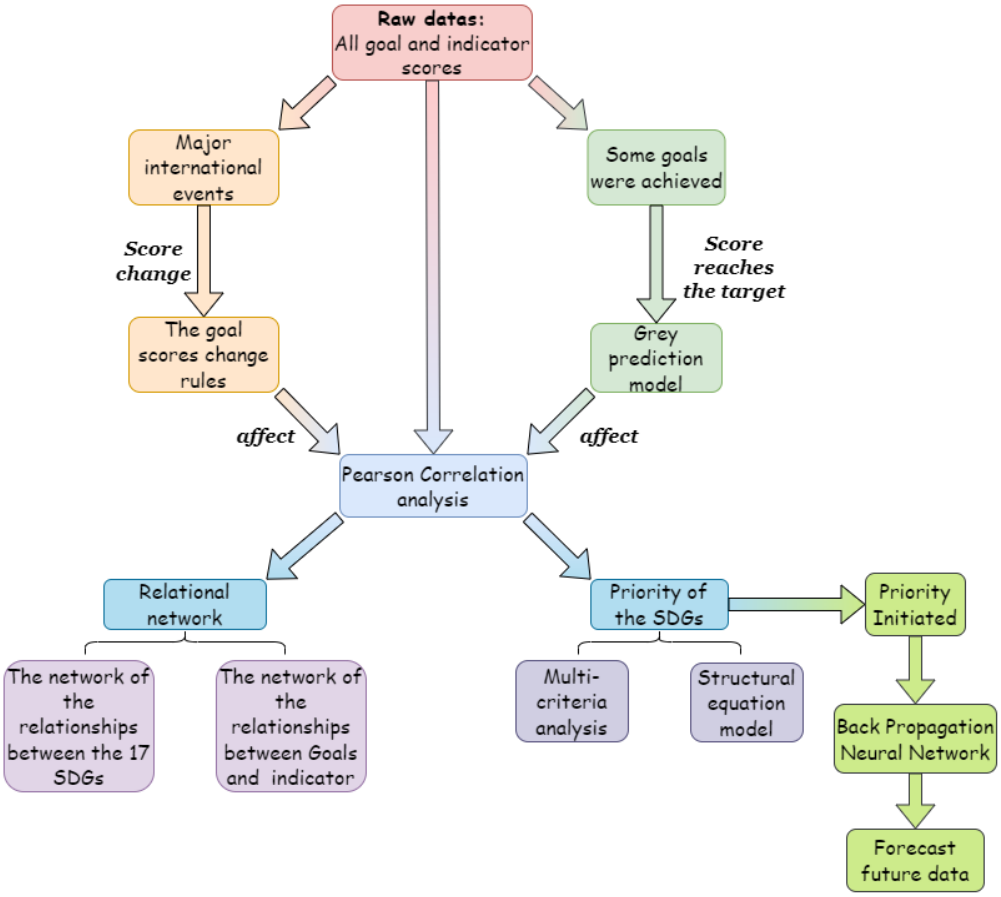

Fig.1 Model Overview

2 Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

- **Assumption 1: The interrelations among sustainable development goals are constrained, and do not account for the secondary and reverse propagation of impacts.**
- **Assumption 2: The influence of diverse factors such as population, geographic environment, and political system across different nations or regions on the sustainable development goals is disregarded.**
- **Assumption 3: The accuracy of scores for different indicators obtained from raw statistical data is precise.**
- **Assumption 4: All the data utilized in this research are genuine and precise.**
- **Assumption 5: The errors of all model outcomes presented in this study are deemed acceptable.**

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
ρ	Quotient of covariance and standard deviation between variables	1
r	Pearson correlation coefficient	2
X_i	Sample i	3
P	SEM path significance coefficient	4
β	SEM weight coefficient of main level	5
λ	SEM path coefficient of main level	6
μ	SEM overall impact path coefficient of hierarchy	7

4 Requirement 1:

This section presents an analysis of Pearson correlation coefficients among the 17 sustainable development goals, leveraging the SDG Index Score as a basis. The resultant correlations are subsequently utilized to construct relationship networks reflective of the interdependent nature of these goals.

4.1 Data Description

The dataset utilized in this study is obtained from the online database for the Sustainable Development Report 2022, which is inclusive of the SDG Index Score. The SDG Index Score quantitatively evaluates the progress of 17 sustainable development goals, alongside multiple specific sub-indicators, within countries worldwide. The assessment is based on a percentage-based scoring system, encompassing the time period between 2000 to 2021.

4.2 Pearson correlation analysis among the SDGs

The Pearson correlation coefficient is a statistical measure of the linear correlation between two variables. An examination of the Pearson correlation coefficients between various SDGs can provide insight into the extent to which they are related to one another.

The Pearson correlation coefficient represents the strength of the linear association between two variables, and is calculated as the covariance of the variables divided by the product of their standard deviations.

$$\rho_{X,Y} = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y} \tag{1}$$

The estimation of the covariance and standard deviation for a given sample population can be used to derive the Pearson correlation coefficient, which is denoted by the lowercase English letter "r".

$$r = \frac{\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i-\bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i-\bar{Y})^2}}. \tag{2}$$

The Pearson correlation coefficient "r" can also be estimated from the mean of the standardized scores of the sample points(X_i, Y_i), resulting in an equivalent expression to the one given above:

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma_X}\right)\left(\frac{Y_i-\bar{Y}}{\sigma_Y}\right). \tag{3}$$

The standardized score, sample mean, and sample standard deviation of the sample points X_i are respectively represented by $\frac{X_i-\bar{X}}{\sigma_X}$, $\bar{X}$, and σ_X.

An analysis was conducted on the 2021 Sustainable Development Goal Index scores of countries worldwide, and a heatmap of the correlation coefficients was generated from the results. The outcomes of this study are presented below:

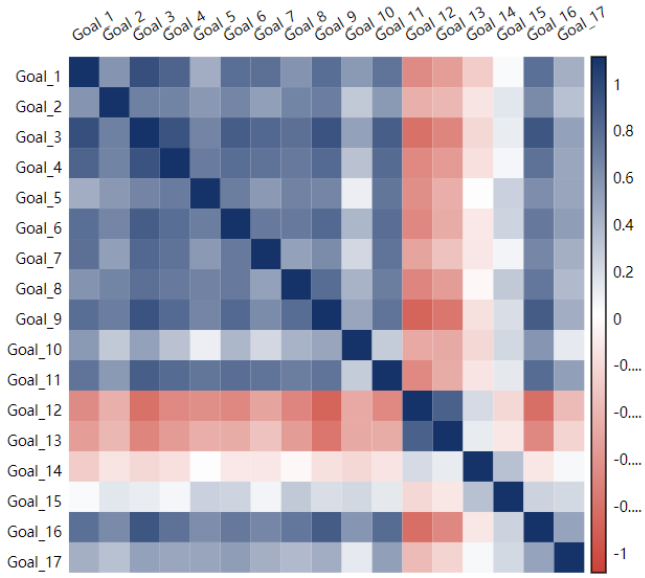

Fig.2 2021 SDGs Thermal diagram of correlation coefficient

4.3 Relationship network construction based on correlation coefficient

The interrelation between Sustainable Development Goals (SDGs) can be segregated into two categories, namely, synergy and trade-off, as reported in the literature. Synergy among SDGs suggests a mutually reinforcing relationship between two or more goals, such that the attainment of one objective facilitates the attainment of the others. On the other hand, trade-offs in SDGs indicate a conflicting relationship between two or more goals, where the accomplishment of one objective leads to a reduction in the achievement of others, and the trade-offs among goals are frequently employed.

The magnitude of the Pearson correlation coefficient (r) is indicative of the correlation between the independent and dependent variables. We have selected a threshold value of 0.6

for r, where a value of 0.6 < r < 1 is deemed as evidence of a synergistic relationship between the two targets. Alternatively, a range of -0.6 < r < 0.6 indicates no significant relationship between the two targets. Finally, when r falls between -1 and -0.6, it suggests a trade-off relationship between the two targets.

According to the calculated results, we can draw a correlation network among the 17 targets.

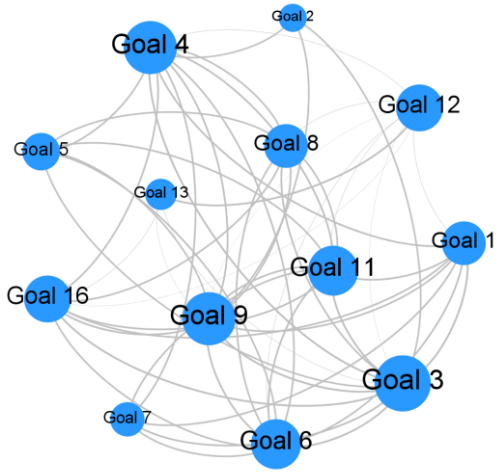

Fig.3 The network of the relationships between the 17 SDGs

The United Nations has put forward sub-indicators for evaluating the advancement towards Sustainable Development Goals (SDGs) under each objective. These sub-indicators are assessed collectively, and a correlation network is constructed to provide a comprehensive understanding of the interconnections among the indicators.

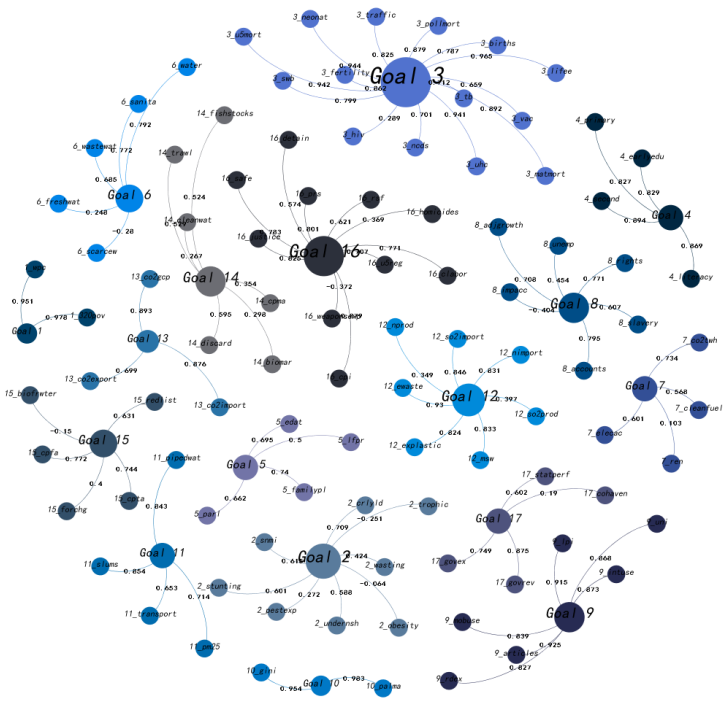

Fig.4 The network of the relationships between the 17 SDGs and indicators

5 Requirement 2:

5.1 SEM structural equation

5.1.1 SEM model building

The evaluation criteria for the 17 Sustainable Development Goals (SDGs) remain unclear, leaving the determination of their degree of accomplishment dependent on the specific indicators included. In this study, we treat the 17 SDGs as endogenous latent variables at the main level, while the specific indicators included under each SDG are considered observed variables at the subordinate level. The analysis includes a total of 120 specific indicators across all 17 SDGs, with 95 indicators utilized as observed variables in the structural equation modeling (SEM) analysis, after removing those indicators that lacked data.

We constructed a hypothetical measurement model of the Sustainable Development Goals (SDG) structural equation using SPSS software. The model was then calibrated using data from 194 countries and regions, including both the overall sustainable development scores and 95 sub-indicators. Through multiple iterations of fitting and refinement, we arrived at a final version of the SDG structural equation, which is presented in Figure.

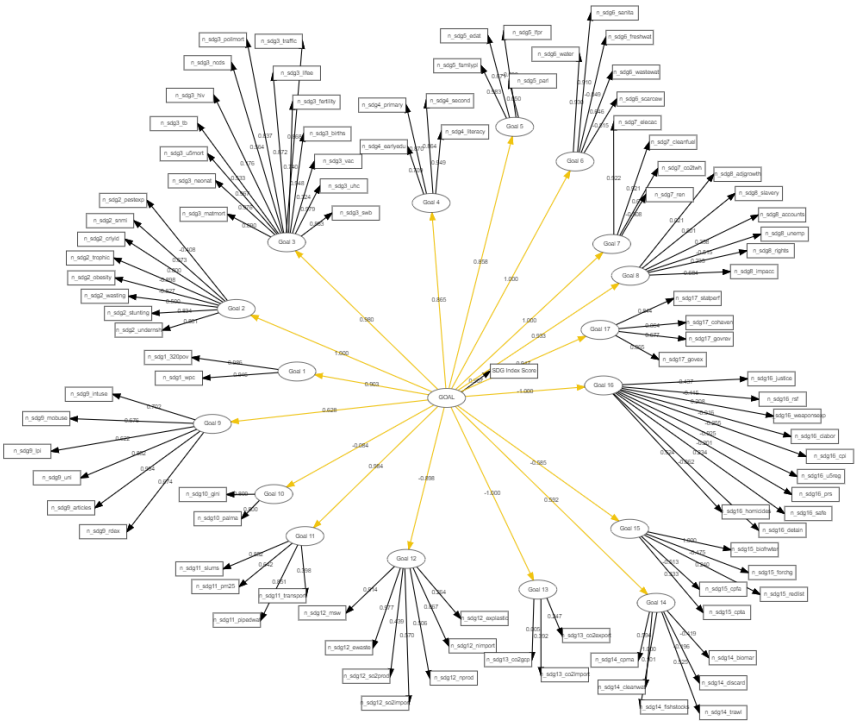

Fig.5 Path coefficient diagram of SEM model SDGs and indicators

In the main hierarchy of our model, the p-significance coefficients for the paths in SDG8, SDG10, SDG13, SDG14, and SDG15 are all greater than 0.05, indicating that these five path relationships cannot be upheld. However, the remaining 13 paths at the main level show significant effects. Specifically, SDG12 and SDG16 exhibit significant negative correlation effects on sustainability as a whole, while SDG1, SDG2, SDG3, SDG4, SDG5, SDG6, SDG7, SDG9, and SDG11 demonstrate significant positive correlation effects on sustainability. At the subor-

dinate level, the observed variables show mostly positive correlation effects on their corresponding main level.

5.1.2 Establishment of the priorities

In structural equation modeling (SEM), factor path loadings measure the strength of the relationship between variables, and larger path loadings indicate higher influence status of the corresponding variable. Based on the calculated path coefficients, we computed the weight coefficients of each transformation content, and then derived the weight coefficients for the main level using equation (4).

$$\beta_i = \frac{\lambda_i}{\sum_{i=1}^{n} \lambda_i} \tag{4}$$

In the equation, the weight coefficient of the main level is denoted by β_i, and the path coefficient of the main level is represented by λ_i.

The calculation of the weight coefficients for the subordinate level can be found in equation (5).

$$\begin{cases} \mu_i = \lambda_i \times \lambda_{ij} \\ \gamma_i = \dfrac{\mu_j}{\sum_{j=1}^{n} \mu_j} \end{cases} \tag{5}$$

In the above equation, μ_i represents the path coefficient of the impact of the subordinate level on overall sustainable development, while λ_{ij} denotes the path coefficient of the subordinate level. Additionally, γ_i represents the weight of the impact of the subordinate level on overall sustainable development.

Taking the "eradication of poverty" at the main level as an example, the process of determining its weight on overall sustainable development is as follows:

Weighting of "GOAL 1: No Poverty":

$$= \frac{0.903}{0.903 + 1 + 1 + 0.98 + 0.865 + 0.858 + 1 + 1 + 0.628 + 0.984 + 0.898 + 1 + 0.917} = 0.082$$

The process of determining the weights of the other main hierarchies follows a similar approach, and they are ranked based on their respective weights. In the case of a positive path coefficient for a main hierarchy, the higher its weight, the higher its priority, while for a negative path coefficient, the higher its weight, the lower its priority. If a main level's path does not hold, it is ranked between the main level with a positive path coefficient and the main level with a negative path coefficient, and the final priority result is presented in the table.

Table2 The priority of SDGs by SEM model

Main level	Weight	Rank
GOAL 1: No Poverty	0.082	5
GOAL 2: Zero Hunger	0.09	1

GOAL 3: Good Health and Well-being	0.089	3
GOAL 4: Quality Education	0.078	6
GOAL 5: Gender Equality	0.077	7
GOAL 6: Clean Water and Sanitation	0.09	1
GOAL 7: Affordable and Clean Energy	0.09	1
GOAL 8: Decent Work and Economic Growth	0	9
GOAL 9: Industry, Innovation and Infrastructure	0.057	8
GOAL 10: Reduced Inequality	0	9
GOAL 11: Sustainable Cities and Communities	0.089	2
GOAL 12: Responsible Consumption and Production	0.081	10
GOAL 13: Climate Action	0	9
GOAL 14: Life Below Water	0	9
GOAL 15: Life on Land	0	9
GOAL 16: Peace and Justice Strong Institutions	0.09	11
GOAL 17: Partnerships to achieve the Goal	0.083	4

5.2 Multi- criteria analytical decision framework

Multi-criteria decision making (MCA) involves the utilization of available decision-related information to rank or give priority to a group of options by manipulating their order. To evaluate the priority of Sustainable Development Goals (SDGs), we devised a decision-making framework for multi-criteria analysis, inspired by the SDG dataset and a prior research by Allen et al. on the ranking of SDGs in Arab nations. The framework incorporates the priority of SDGs based on two dimensions, namely the "urgency" and the "systemic impact" of the SDGs, as a means of integrating the SDG priorities.

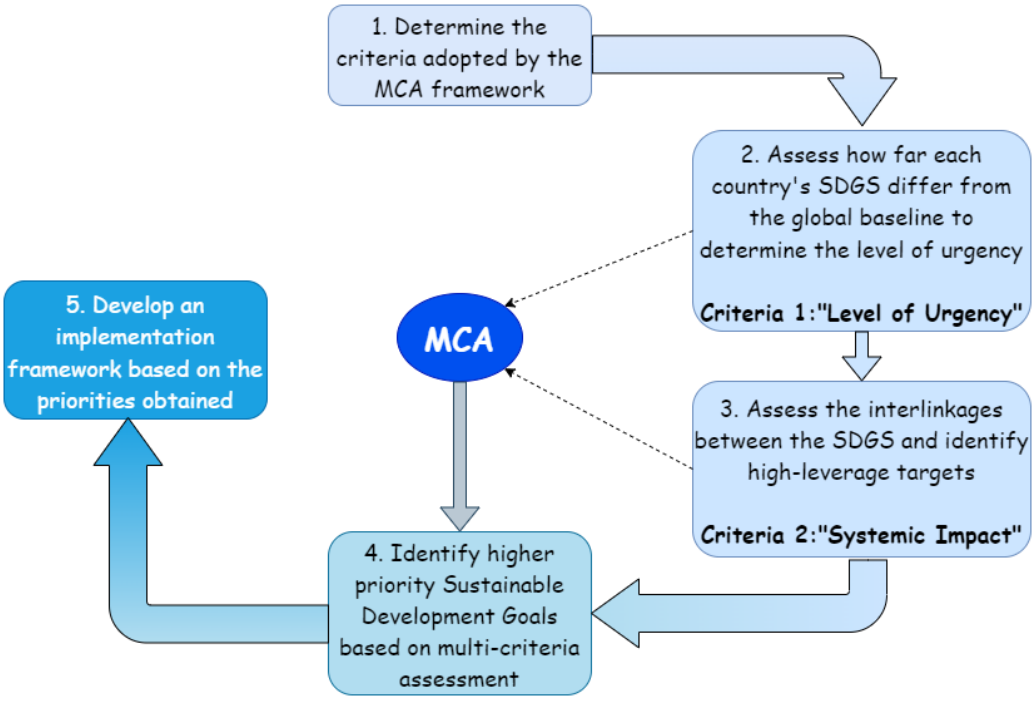

Fig.6 MCA Frame

Criterion 1: Level of urgency: baseline assessment.

In this study, a baseline assessment was conducted to evaluate the current value of a country's sustainable development goals (SDGs) relative to the global SDG baseline and average, using the SDR dataset. The degree of progress towards SDGs for each country was assessed and categorized into five levels, providing a means to evaluate the urgency of each goal and assign a corresponding score.

Table3 The "urgency" assessment category for each SDG indicator

Category and score	Rule Applied
5	Goal Achievement
4	Challenges remain
3	Significant challenges
2	Major challenges
1	Insufficient data

Criterion 2: Systemic impact: interlinkages among the SDGs.

This stage involves determining the SDGs that exhibit a greater degree of "systemic impact" compared to other SDGs. A semi-quantitative cross-impact matrix assessment is employed, which involves analyzing the interconnections between the goals via a cross-impact matrix analysis. Based on the interactions between the SDGs established in Requirement 1, a seven-point scale was employed to assign scores to the seven levels, which were derived by multiplying the correlation coefficients by a specific factor. A positive score denotes a synergistic impact, a negative score denotes a trade-off impact, and a score of 0 signifies no impact between the SDGs.

Table3 Score the impact of SDG goal on another goal

Name	Explanation
Indivisible	Inextricably linked to the achievement of another target
Reinforing	Aids the achievement of another target
Enabling	Creates conditions that further another target
Consistent	No significant positive or negative intreractions(neutral)
Constraining	Limits options on another target
Counteracting	Clashes with another target
Cancelling	Makes it impossible to reach another target

Normalize the obtained urgency score and systemic impact score using the following formula:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{6}$$

To assign weights for urgency and systemic impact, a ratio of 1:1 is assumed. Subsequently, the priority ranking of the total score can be derived.

Table4 The priority ranking of the total score

SDG	urgency degree	Urgent level priority	Systematic influence	The priority of system impact	Total score	Priority
1	1.000000	1	1.000000	2	0.635690236	3

2	0.845118	10	0.948148	6	0.6503367	2
3	0.781145	11	0.898148	1	0.776094276	1
4	0.720539	12	0.627778	3	0.556818182	6
5	0.700337	13	0.587037	5	0.529208754	8
6	0.649832	14	0.455556	4	0.598400673	4
7	0.589226	15	0.368519	8	0.317340067	14
8	0.585859	16	0.240741	7	0.534427609	7
9	0.569024	17	0.133333	16	0.347390572	13
10	0.552189	2	0.118519	11	0.382323232	11
11	0.491582	3	0.114815	9	0.312457912	15
12	0.471380	4	0.114815	10	0.126599327	16
13	0.393939	5	0.114815	17	0	17
14	0.323232	6	0.114815	12	0.557407407	5
15	0.215488	7	0.114815	13	0.447979798	9
16	0.134680	8	0.105556	14	0.417676768	10
17	0.000000	9	0.000000	15	0.3503367	12

5.3 A 10-year outlook after the launch of the priorities

5.3.1 Deep neural network prediction model

Given the intricate interdependencies that exist among diverse sustainable development objectives, it is often challenging for conventional prediction models to account for the collective influence of these goals. In light of this, the current study employs deep neural networks to forecast the likely outcomes of sustainable development goals ten years into the future.

The deep neural network prediction model is composed of an input layer, output layer, and hidden layers. Specifically, the input layer comprises the scores of 17 sustainable development goals from 10 years prior, while the output layer consists of the same 17 goals' scores from 10 years later. By collating the scores for the 17 sustainable development goals across 194 countries and regions from 2000 to 2021, a total of 2,124 training samples were obtained. Among these, 70% were designated for training, and 30% for testing. The model's training performance is depicted in the figure, where the red pentagram represents the predicted results of the test set.

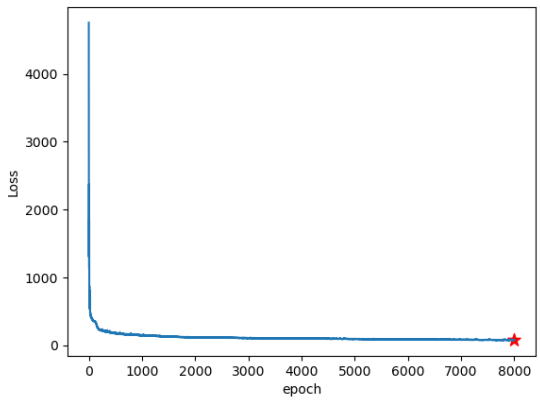

Fig.7 The diagram of deep neural network model training loss

The figure illustrates that the red pentagram is located at the tail end of the Loss curve, which suggests that the prediction error of the test set is in agreement with that of the training set. This outcome indicates that the neural network has successfully captured all the characteristics of the sample data, and thereby confirms the accuracy and effectiveness of the proposed prediction model.

5.3.2 Forecast for 10 years after the initiation of priorities

Based on a deep neural network prediction model, this study employed the sustainable development goal scores of the world in 2021 as a benchmark to forecast the scores in the following 10 years. Subsequently, the priority action was established, which aimed to increase the target scores by 20% higher than this benchmark. The pre-trained deep neural network prediction model was then applied to predict the scores of various indicators 10 years later.

Table5 Prediction of SDG scores after 10 years based on deep neural network

SDG	Initial score 1	Predicted score 1	Initial score 2	Predicted score 2	Initial score 3	Predicted score 3
1	73.1	87.4	73.1	95.6	73.1	90.8
2	64.4	69.3	77.3	84.3	64.4	72.4
3	69.4	81.1	69.4	86.8	83.2	94.6
4	80.2	91.9	80.2	98.3	80.2	89.6
5	59.0	68.1	59.0	74.9	59.0	69.3
6	65.4	71.3	78.5	82.6	65.4	69.5
7	67.4	76.2	80.9	86.4	67.4	78.1
8	68.2	73.9	68.2	77.4	68.2	74.2
9	60.1	66.4	60.1	76.2	60.1	69.5
10	36.8	42.2	36.8	59.7	36.8	48.1
11	66.4	77.2	66.4	85.7	66.4	78.1
12	88.8	89.3	88.8	94.8	88.8	92.3
13	86.0	90.5	86.0	92.1	86.0	91.0
14	59.5	65.3	59.5	67.9	59.5	65.8
15	58.1	63.9	58.1	65.4	58.1	64.6
16	64.3	69.7	64.3	79.8	64.3	72.0
17	55.8	65.1	55.8	67.9	55.8	65.7

The present study utilizes a deep neural network prediction model to estimate the sustainable development goal (SDG) scores for the world in the next decade. Specifically, the benchmark score for SDGs in 2021 serves as the initial score 1, from which the score 10 years later is predicted. Additionally, two priority action scenarios are considered based on the SEM and MCA models, respectively, in which the target scores for certain SDGs are raised by 20% above the initial scores. The predicted scores for these two scenarios, denoted as scores 2 and 3, are also obtained from the deep neural network prediction model.

The performance of the SDGs in the future is evaluated based on a historical data analysis and a target threshold of 85 points. The results obtained from the prediction model suggest that, without any priority action, SDG1, SDG4, SDG12, and SDG13 will be achieved. On the other hand, when the SEM model-based priority action is initiated, SDG1, SDG3, SDG4, SDG7,

SDG11, SDG12, and SDG13 will be realized. Meanwhile, the MCA-based priority action leads to the achievement of SDG1, SDG3, SDG4, SDG12, and SDG13.

6 Requirement 3:

This section provides an analysis of the effects on the sustainable development goal network and priorities obtained in the preceding section upon the attainment of a specific goal, characterized by a score of 85, in the context of ongoing development. In addition, recommendations for additional goals to be presented to the United Nations are put forth.

6.1 Post-year target score prediction based on grey prediction model

To get data on all target scores after a certain goal is achieved, it is necessary to forecast the data after several years of development, and here we introduce a gray prediction model.

Gray forecasting is a simple time series forecasting model, suitable for fitting and forecasting based on less data, by identifying the degree of dissimilarity of development trends among system factors, i.e., correlation analysis, and generating processing of raw data to find the pattern of system changes, generating data series with strong regularity, and then establishing the corresponding differential equation model to predict the future development trend of things.

Fundamental: The GM(1,1) model is the core of gray forecasting, which is a first-order differential equation model for individual variable forecasting with an approximate exponential law of discrete-time response function.The method for building the GM(1,1) model is that:

Let $X^{(0)} = \{X^{(0)}(1), X^{(0)}(2), \dots\dots, X^{(0)}(n)\}$be the original non-negative time series and $X^{(1)}(t)$ be the cumulative production series,i.e.$X^{(0)}(t) = \sum_{m=1}^{i} X^{(0)}(m), t = 1,2, \dots\dots n$.

The whitening differential equation for the GM(1,1) model is that:

$$\frac{dX^{(1)}}{dt} + aX^{(1)} = u \tag{7}$$

A is the parameter to be identified,also known as the development coefficient; u is the endogenous variable to be identified, also known as the amount of gray action.Let the vector $\hat{a} = \binom{a}{u}$ to be identified, By least squares method to find that in the $\hat{a} = (B^T B)^{-1} B^T y$:

$$B = \begin{vmatrix} -\frac{1}{2}\left(X^{(1)}(1) + X^{(1)}(2)\right) & 1 \\ -\frac{1}{2}\left(X^{(1)}(2) + X^{(1)}(3)\right. & 1 \\ \dots\dots. & \dots. \\ -\frac{1}{2}\left(X^{(1)}(n-1) + X^{(1)}(n)\right. & 1 \end{vmatrix}; \ y = \begin{vmatrix} X^{(0)}(2) \\ X^{(0)}(3) \\ \dots \\ X^{(0)}(n) \end{vmatrix} \tag{8}$$

The discrete time response function for the gray prediction can then be obtained as:

$$X^{(1)}(t+1) = \left(X^{(0)}(1) - \frac{u}{a}\right) e^{-at} + \frac{u}{a} \tag{9}$$

The resulting from $X^{(1)}(t+1)$ cumulative predicted values are reduced to the predicted values as

$$\hat{X}^{(0)}(t+1) = \hat{X}^{(1)}(t+1) - \hat{X}^{(1)}(t),\ (t = 1、2、3 \cdots\cdots n) \tag{10}$$

The flow chart is as follows:

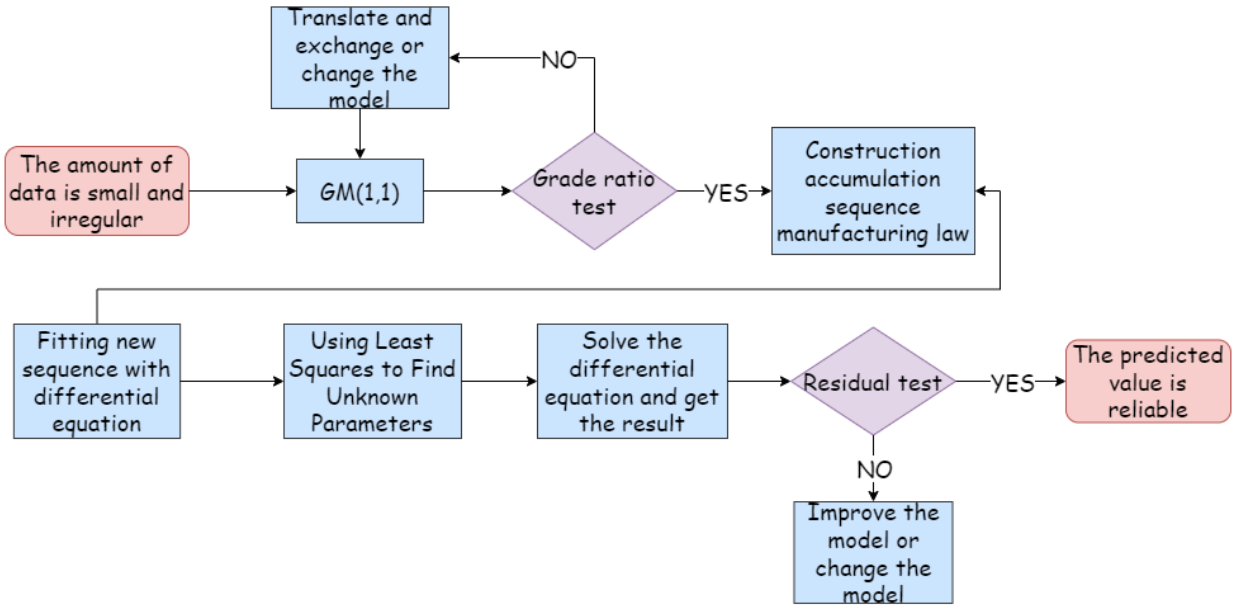

Fig.8 Flow chart of grey prediction model

Before building the gray prediction model GM(1,1), the time series is tested by the cascade test. If the series passes the rank-ratio test, the series is suitable for building the gray model, and if it does not pass the rank-ratio test, the series is "transformed" so that the new series meets the rank-ratio test. After the prediction, the gray prediction model has to be tested to determine whether it is reasonable, and only the model that passes the test can be used for prediction. The system is mainly used to test the gray forecasting model by the posterior difference ratio C-value. The data of 102 countries are tested by the rank ratio test and the posterior difference ratio C-value test, and the data of 70 countries are finally selected from the results.

After 10 years (2030) several countries have scored 85 on Goal 4 and can be considered to have achieved the goal in 2030.

6.2 The changes of network and priorities

Based on the predicted data, using the previous model, it can be concluded that the overall correlation coefficient and relationship network becomes, after the achievement of objective 4:

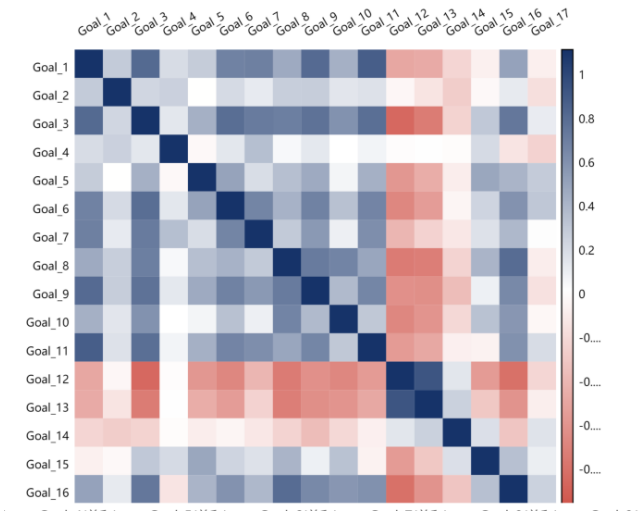

Fig.9 SDGs Thermal diagram of correlation coefficient after goal 4 is achieved

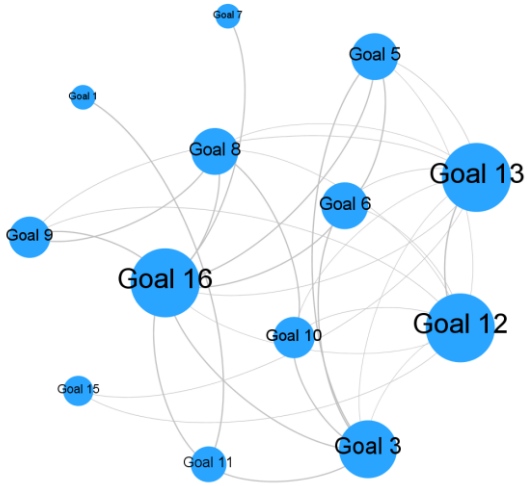

Fig.10 The network of the relationships between the 17 SDGs after goal 4 is achieved

Priority becomes:

Table6 SDGs priority changes after goal 4 is achieved

Original priority ranking	priority ranking after goal 4 is achieved
SDG3	SDG3(Unchanged)
SDG 2	SDG1(Up 1 place)
SDG 1	SDG11(Up 12 place)
SDG 6	SDG7(Up 10 place)
SDG 14	SDG8(Up 2 place)
SDG 4	SDG5(Up 2 place)
SDG 8	SDG2(Down 5 place)
SDG 5	SDG14(Down 3 place)
SDG 15	SDG15(Unchanged)
SDG 16	SDG16(Unchanged)
SDG 10	SDG17(Up 1 place)
SDG 17	SDG9(Up 1 place)
SDG 9	SDG12(Up 3 place)
SDG 7	SDG6(Down 10 place)
SDG 11	SDG13(Up 2 place)
SDG 12	SDG10(Down 5 place)
SDG 13	

6.3 Other goals to consider

1. Convenient Transportation System

In the agenda for sustainable development goals, special attention is given to the issue of inequality. Inequality is related to many key development issues, such as income, justice, ex-

pression rights, job opportunities, education, health, and so on. In the transportation field, inequality is also a critical issue. Among the 17 sustainable development goals that have been proposed, transportation-related issues are distributed among five goals and indicators: indicator 3.6, related to road safety; specific goal 7.3, related to energy efficiency; goal 9.1 related to sustainable infrastructure; specific goal 11.2, related to urban transportation; and goal 12.C, related to fossil fuel subsidies. These existing indicators and goals cannot fully represent the important role of the transportation system in sustainable development.

As global connections deepen, the ability of transportation systems to provide people with opportunities is unequal in many cities, regions, and countries around the world. Transportation planning is shifting from providing infrastructure/mobility to ensuring people's ability to reach their destinations. Therefore, the focus is shifting towards promoting sustainable development through multimodal integration (providing integration of multiple modes, promoting the use of walking, cycling, and public transport, and ensuring a good combination of transportation modes) and promoting transportation equity.

2. Cultural Exchange

In recent decades, culture has received increasing attention as a mediator or driving force for sustainable development. In 2005, the United Nations Educational, Scientific and Cultural Organization (UNESCO) adopted the Convention on the Protection and Promotion of the Diversity of Cultural Expressions, which raised awareness of the role of culture in sustainable development.

On the one hand, culture can affect the achievement of goals related to physical and mental health. Culture influences physical health by promoting or hindering disease prevention, detection, and treatment. On the other hand, the role of culture in justice and equality has been widely studied in various disciplines, including education, gender sociology, economics, politics, and criminology. Additionally, actors in economic systems find it difficult to escape the influence of cultural environments. All of this indicates that culture and sustainable development are inseparable.

Despite the effective existing analysis and evidence of the relationship between culture and sustainable development, two barriers prevent a deeper understanding of the connection between the two. First, it is difficult to compare the impact of specific cultural features on the SDG's specific goals. Second, research resources are unevenly distributed among cultural features and do not address how the relationship between culture and sustainable development goals varies depending on cultural characteristics.

7 Requirement 4:

This section describes how changes in the international environment, such as technological advances and global epidemics, affect the entire network of relationships by influencing scores on certain goals, and affect the process of achieving the UN's goals.

7.1 Rules and data on the impact of specific changes on the SDG

Measuring the impact of unpredictable factors, such as technological advancements, global pandemics, and climate change, on the Sustainable Development Goals (SDGs) poses a

significant challenge for quantification. To better determine the effects of a particular change on the SDGs, a two-step process is proposed. Firstly, the impact of the change on the individual indicators contained within the SDGs is analyzed, and secondly, the total impact on the SDG is summarized based on the affected indicators.

In order to assess the impact of specific changes on the SDGs, a set of impact rules is proposed. A specific change is considered to have an impact on a small indicator if it affects it, and no impact if it does not affect it. The number of affected small indicators is then tallied for each SDG. If the proportion of affected small indicators is less than 30% of the total, the impact is considered low, if it falls between 30% and 60%, it is moderate, and if it exceeds 60%, it is considered high.

Drawing on historical data and real-life scenarios, a set of change rates is proposed for the impacts of technological advancements, climate change, and regional conflicts. Specifically, for technological advancements, low, moderate, and high impacts correspond to change rates of +2%, +4%, and +6%, respectively. For climate change, low, moderate, and high impacts correspond to change rates of -1%, -3%, and -5%, respectively. For regional conflicts, low, moderate, and high impacts correspond to change rates of -3%, -6%, and -9%, respectively. Considering the differences in the global impact of these changes, it is assumed that 50%, 100%, and 25% of countries globally are affected by technological advancements, climate change, and regional conflicts, respectively.

Table7 The rule of influence by specific changes

Specific changes	**Impact amplitude**			**Scope of influence**
	low	moderate	high	
Technological advancements	+2%	+4%	+6%	50%
Climate change	-1%	-3%	-5%	100%
Regional conflicts	-3%	-6%	-9%	25%

According to the above rules, the final impact magnitude of technological advancements, climate change, and regional conflicts on the 17 Sustainable Development Goals (SDGs) is shown in the table.

Table8 The magnitude of the impact of a particular change on the SDGs

SDG	Technological advancements	Climate change	Regional conflicts
1	-	-1%	-6%
2	+2%	-3%	-6%
3	+4%	-	-9%
4	+2%	-	-6%
5	+2%	-	-
6	+4%	-	-3%
7	+6%	-	-9%
8	+2%	-	-3%

9	+4%	-	-
10	-	-	-6%
11	+4%	-3%	-9%
12	-	-	-
13	-	-5%	-
14	-	-	-
15	-	-	-
16	+4%	-	-9%
17	-	-	-3%

7.2 The changes of network and priorities

Based on the data after the impact, using the model mentioned earlier, it can be concluded that a certain degree of technological progress will cause the overall relationship network to change to:

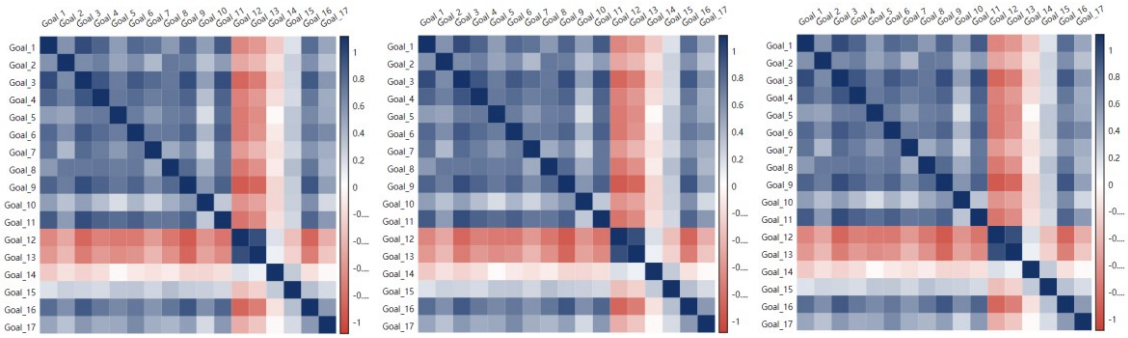

Fig.11 Thermal diagram of correlation coefficient(Technological progress, Climate change, Regional wars)

Priority becomes：

Table9 The priority of the SDGs changes after specific change

Original		Technological advancements		Climate change		Regional conflicts	
SDG	**Score**	**SDG**	**Score**	**SDG**	**Score**	**SDG**	**Score**
SDG3	4.762	SDG3	4.724	SDG3	4.72	SDG3	4.725
SDG1	4.513	SDG1	4.702	SDG1	4.696	SDG1	4.697
SDG4	4.225	SDG4	4.202	SDG4	4.198	SDG4	4.203
SDG6	2.777	SDG6	2.799	SDG6	2.791	SDG6	2.788
SDG5	2.549	SDG5	1.945	SDG5	1.945	SDG5	1.929
SDG2	1.853	SDG2	1.911	SDG2	1.91	SDG2	1.919
SDG8	1.374	SDG8	1.358	SDG8	1.357	SDG8	1.374
SDG7	0.684	SDG7	0.664	SDG7	0.666	SDG7	0.661
SDG11	0.099	SDG12	0.061	SDG12	0.061	SDG12	0.067
SDG12	0.016	SDG10	0	SDG10	0	SDG10	0
SDG10	0	SDG14	0	SDG14	0	SDG14	0
SDG14	0	SDG15	0	SDG15	0	SDG15	0
SDG15	0	SDG16	0	SDG16	0	SDG16	0
SDG16	0	SDG17	0	SDG17	0	SDG17	0
SDG17	0	SDG9	-0.06	SDG9	-0.063	SDG9	-0.068

SDG9	-0.056	SDG11	-0.584	SDG11	-0.587	SDG11	-0.573
SDG13	-0.624	SDG13	-0.679	SDG13	-0.679	SDG13	-0.67

7.3 The United Nations development process

According to the above analysis, major international events such as technological progress, climate warming and regional wars have almost no impact on the prioritization of SDGs within a certain range, and the network of relationships fluctuates less, but they will affect the correlation coefficient of each goal, thus affecting the degree of impact of the goal on the whole system.

Based on this, the United Nations should adopt appropriate governance measures or policies for the goals that are more negatively affected while keeping the overall development process unchanged; it should maintain constant attention to the goals that are positively affected and take the opportunity to fully develop and achieve the goal at an early date.

8 Requirement 5:

To benefit from the opportunities and challenges presented by the SDGs, companies and organizations should identify their priorities to focus on key areas.

We were inspired by the United Nations SDG Compass Guide to develop a three-step framework for how companies and organizations can identify their priorities.

Step 1: Analyze the company or organization's value chain to determine areas of impact.

Step 2: Select indicators and collect data.

Step 3: Use the MCA framework from step 2 to determine priorities.

Step 1：Analyze the company's value chain to determine areas of impact.

Companies or organizations should evaluate the positive or negative impacts of their business on the SDGs throughout the entire value chain, starting with the entire value chain from the supply base to inbound logistics, from production, operations to distribution, use, and disposal of products, as the first step of the overall framework. Companies do not need to assess all SDGs in all parts of the value chain in detail, but rather focus on the areas with the greatest impact. For example, if a company is engaged in a labor-intensive industry or the region where its supply chain is located has low wages, poor labor rights and standards enforcement, that part may be identified as a "potentially high impact" area.

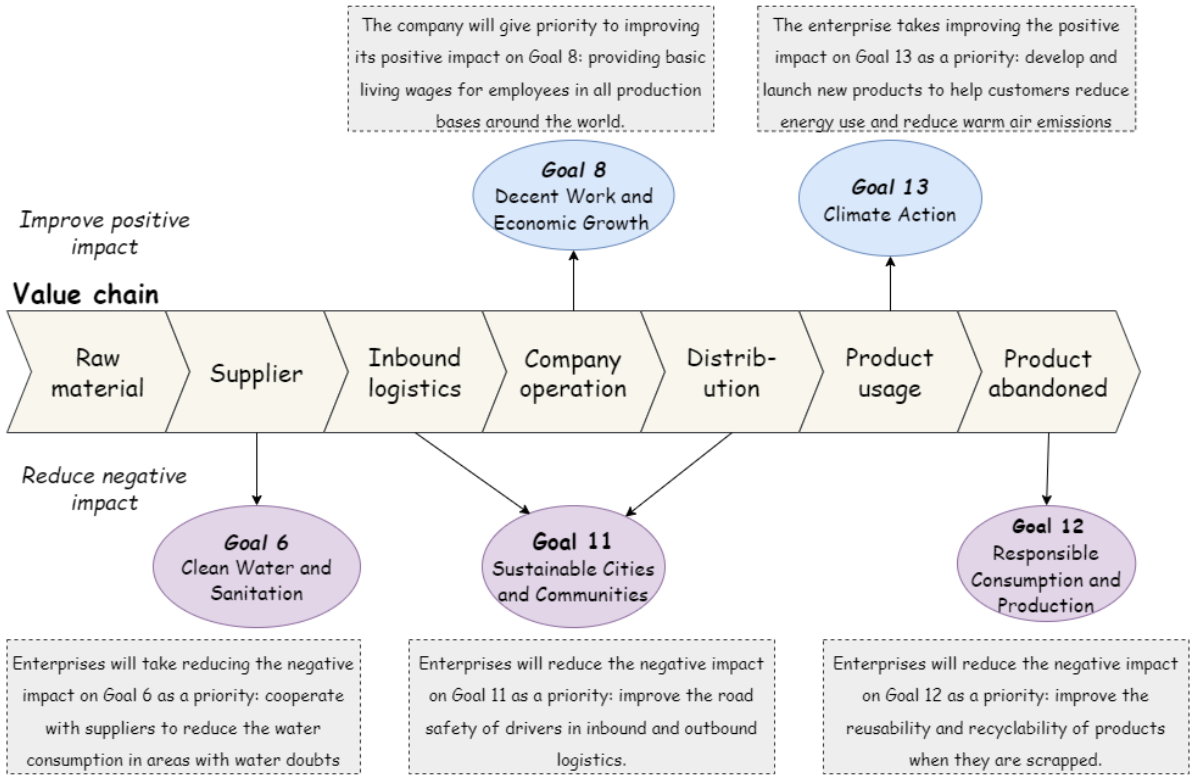

Fig.12 Mapping SDGs on the value chain

Step 2：Selecting metrics and collecting data

Companies and organizations can use Logic model to analyze a five-step process, tracking the path of the company's activities from inputs, activities, outputs, outcomes, and impacts, and analyzing the results to determine what data should be collected.

The SDG Compass website, www.sdgcompass.org, provides a range of business metrics applicable to the 17 United Nations Sustainable Development Goals and their targets, including existing widely recognized and used metric systems such as the Global Reporting Initiative (GRI) and the Carbon Disclosure Project (CDP) as well as other relevant metric systems. Companies can select the most appropriate metrics for the high impact areas.

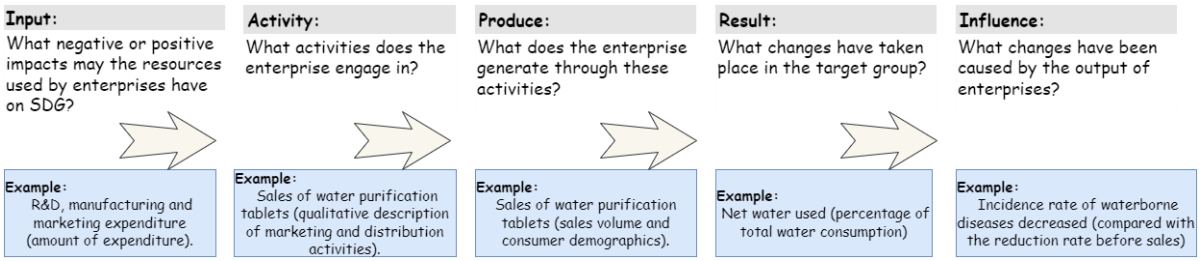

Fig.13 Logic model

Step 3：Setting priorities

After understanding the current and potential negative and positive impacts on sustainable development and collecting data, companies should prioritize SDGs. Companies and organizations can divide their focus areas into different dimensions. For example, similar to the dimensions in Requirement 2, "urgency" and "systemic impact," companies can first score the collected metric data in Step 2 according to their magnitudes, evaluate the urgency of each

SDG based on the difference between the metric and the optimal target. Then, by analyzing the interdependence of each metric, select "high leverage targets" to determine the systemic impact of the SDGs. Finally, by multiplying the scores and weights of different dimensions and adding them up, companies can select their priorities.

It should be noted that in the process of executing the three steps, assessing business impacts and determining priorities is not a scientific process but rather a subjective judgment process. Therefore, companies should maintain transparent records of the process. It is also recommended to repeat these three steps periodically (e.g. annually) to track the development trends of relevant impacts and priorities.

9 Sensitivity Analysis

In order to substantiate the durability of the relational network structure model and MCA framework model for prioritization, an assessment was conducted using the SDG1 score data for the year 2021 from various countries. Specifically, the data was subject to a positive and negative perturbation of 10% to validate the model's stability under varying conditions.

Heat maps of the correlation impact coefficients after a 10% boost and a 10% drop were obtained by the Pearson correlation analysis method of demand one.

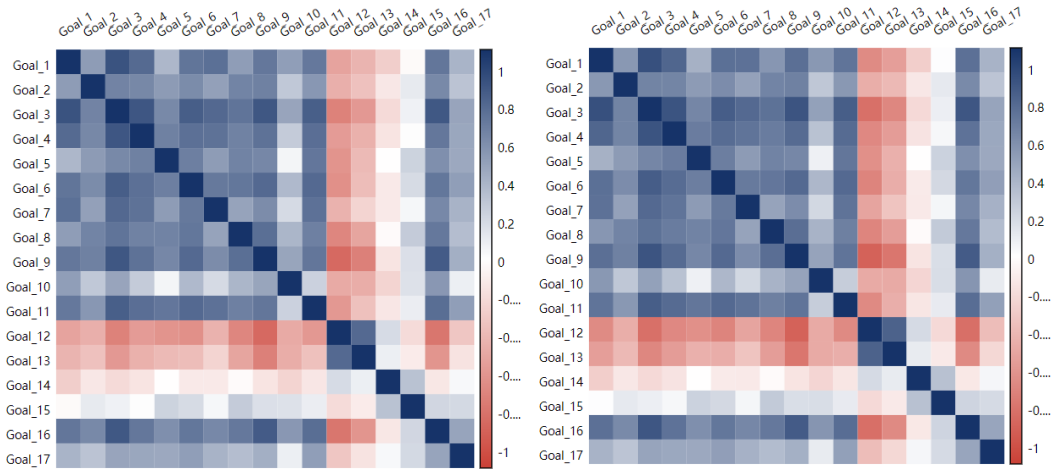

Fig.14 Thermal diagram of correlation coefficient of sensitivity analysis（SDG1 raised by 10%，SDG1 declined by 10%）

The heat map analysis reveals that a slight perturbation in a single target did not elicit noteworthy modifications in the network structure of interdependent sustainable development goals (SDGs). Hence, it can be inferred that the mutual relationships between SDGs remained largely constant, and the overall structure of the relationship network persisted without any significant alterations.

Requirement 2 of the study employed the MCA framework to measure system impact as the primary assessment metric. This approach facilitated the prioritization of SDGs by subjecting the system to a 10% positive and negative variation, respectively. The ranking of SDG priorities was consequently derived from this assessment.

Table10 Priority change of sensitivity analysis

Original priority ranking		The priority ranking after raised by 10%		The priority ranking after declined by 10%	
SDG	**Score**	**SDG**	**Score**	**SDG**	**Score**
SDG3	1.000	SDG1	1.000	SDG1	1.000
SDG1	0.948	SDG4	0.853	SDG4	0.944
SDG4	0.898	SDG3	0.841	SDG3	0.932
SDG6	0.628	SDG6	0.576	SDG6	0.672
SDG5	0.587	SDG5	0.395	SDG5	0.496
SDG2	0.456	SDG2	0.378	SDG2	0.479
SDG8	0.369	SDG8	0.284	SDG8	0.386
SDG7	0.241	SDG7	0.150	SDG7	0.256
SDG11	0.133	SDG11	0.032	SDG11	0.140
SDG12	0.119	SDG12	0.015	SDG12	0.124
SDG10	0.115	SDG10	0.012	SDG10	0.121
SDG14	0.115	SDG14	0.012	SDG14	0.121
SDG15	0.115	SDG15	0.012	SDG15	0.121
SDG16	0.115	SDG16	0.012	SDG16	0.121
SDG17	0.115	SDG17	0.012	SDG17	0.121
SDG9	0.106	SDG13	0.000	SDG9	0.109
SDG13	0.000	SDG9	0.000	SDG13	0.000

The ranking variations depicted in the table reveal that any modifications in SDG1 resulted in a consistent emergence of SDG1 as the top priority. This outcome indicates an augmented influence of SDG1 on the other SDGs, with minimal alterations noted in the ranking of the remaining SDGs. Therefore, the stability of the established MCA framework model in effectively managing the response to perturbations in a single target can be substantiated through this finding.

10 Model Evaluation and Further Discussion

10.1 Strengths

1. It is scientifically reasonable to construct the relationship network between objectives based on correlation coefficients, and the correlation coefficients are obtained based on Pearson correlation analysis to evaluate the correlation between objectives;

2. Comprehensive and objective assessment of target priorities, using both multidimensional MCA and specific metrics-based SEM models to derive priorities for multiple judging criteria;

3. For accurate prediction of future data, for different problems we used two methods, BP neural network and grey prediction model, to predict the data for specific situations;

4. Data visualization is clear and viewable, drawing a variety of visual graphics such as heat map of correlation coefficients, relational network diagrams, curve diagrams, algorithm flow charts, etc. to present data and content intuitively;

5. The sensitivity analysis of the model proved the validity of the model under different combinations of parameters and proved the robustness of the model;

6. Based on real-life specific events, the data is analyzed specifically and reasonable advice is given to the United Nations and companies in the context of the actual situation.

10.2 Weaknesses

1. Some of the targets with small changes in scores also have smaller changes in coefficients using Pearson correlation analysis, and the fluctuations in the relationship network may also be smaller;

2. In some specific cases, the small amount of available valid data may cause errors in the results, and the prediction of data by BP neural networks cannot fully simulate the interaction between objectives;

3. Under this problem, the MCA model can choose only two valid dimensions, which may not be comprehensive enough for ranking the target priorities.

10.3 Further Discussion

1. Refine the relationship network structure to make network change fluctuations more visible;

2. Considering the interaction between the objectives, the BP neural network model can be improved to a graph neural network model to achieve more accurate prediction;

3. Increase the analytical dimensions of the MCA model to achieve a more comprehensive analysis;

4. For the impact of a specific event on the target, you can find the relevant specific impact data, so as to achieve a more objective analysis of the forecast.

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